Finding My Closest Friends Using K-Nearest Neighbours (KNN)

Machine Learning Project Report

Dataset: kNN Project.xlsx
Participants: 19
Questions: 11 (Scale 1–3)

Abstract

This project applies the K-Nearest Neighbours (KNN) algorithm to identify participants whose questionnaire responses are most similar to mine using Euclidean distance. The project includes data collection, KNN modelling, Euclidean distance ranking, bar chart, friendship network graph, correlation heatmap, PCA and K-Means clustering.

Methodology

Workflow:
1. Collect responses using Google Forms.
2. Import dataset with pandas.
3. Select questionnaire features.
4. Train a Nearest Neighbors model using Euclidean distance.
5. Rank participants by similarity.
6. Visualize results using bar chart, network graph, correlation heatmap, PCA and K-Means clustering.

Key Python Code

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neighbors import NearestNeighbors
```

Load Dataset

```
df = pd.read_excel("kNN project.xlsx")
```

Prepare Features

```
X = df.iloc[:,2:]
```

Build KNN Model

```
knn = NearestNeighbors(n_neighbors=len(df), metric="euclidean")
knn.fit(X)
distances, indices = knn.kneighbors(X.iloc[[0]])
```

Create Results Table

```
results=[]
for i in range(len(indices[0])):
    results.append({
        "Rank": i+1,
```

```
        "Name": df.iloc[indices[0][i]]["Name"],
        "Euclidean Distance": round(distances[0][i],2)
    })
results_df = pd.DataFrame(results)
```

Bar Chart

```
plt.bar(results_df["Name"],results_df["Euclidean Distance"])
plt.xticks(rotation=45)
plt.show()
```

Correlation Heatmap

```
questions=df.iloc[:,2:]
corr=questions.corr()
plt.matshow(corr,cmap="coolwarm")
plt.colorbar()
plt.show()
```

PCA

```
from sklearn.decomposition import PCA
pca=PCA(n_components=2)
X_pca=pca.fit_transform(X)
plt.scatter(X_pca[:,0],X_pca[:,1])
```

K-Means

```
from sklearn.cluster import KMeans
kmeans=KMeans(n_clusters=3,random_state=42,n_init=10)
clusters=kmeans.fit_predict(X)
```

Network Graph

```
import networkx as nx
G=nx.Graph()
# Add nodes, edges, colours and labels as implemented in the project.
```

Results & Interpretation

• Obiwuru had the smallest Euclidean distance and was the closest match.
• Participants with smaller Euclidean distances were more similar to the author's responses.
• PCA revealed natural spatial groupings.
• K-Means grouped participants into three clusters.
• The network graph visually represented participant similarity.

Visualizations

1. Euclidean Distance Bar Chart

Interpretation:

This chart ranks participants based on their similarity to my responses. Shorter bars indicate greater similarity, while taller bars indicate less similarity

2. Friendship Network Graph

Interpretation:

This graph shows the relationship between my responses and other participants. The colours represent different similarity levels, while the numbers on the connecting lines show the Euclidean distance

3. Correlation Heatmap

Interpretation:

This heatmap shows how the questionnaire questions relate to one another. Red indicates a stronger positive relationship, while blue indicates a weaker or negative relationship.

4. PCA Visualization

Interpretation:

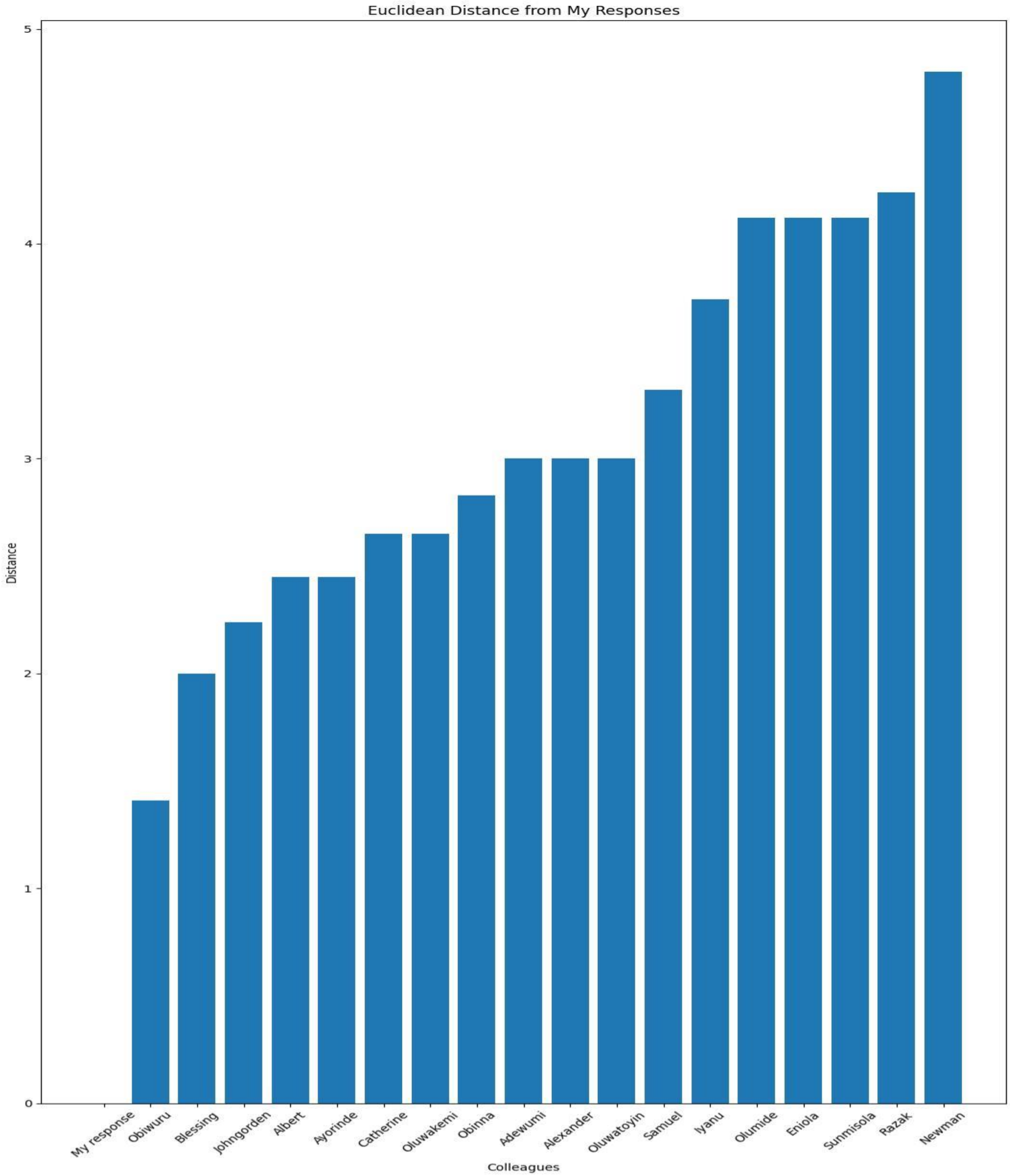
The PCA plot reduces the 11 questionnaire responses into two dimensions. Participants closer together have more similar response patterns, while those farther apart are less similar.

5. K-Means Clustering

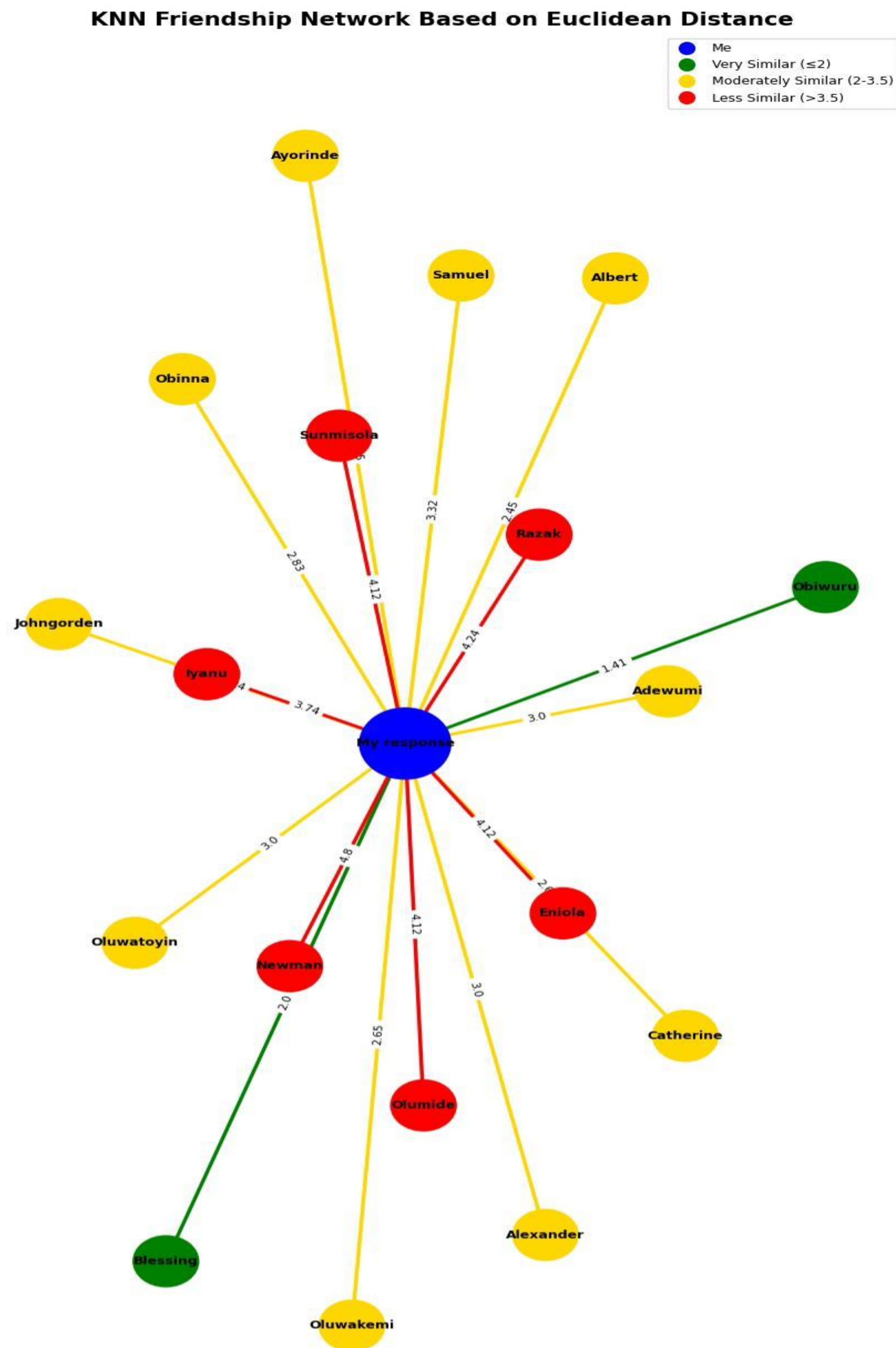
Interpretation:

This visualization groups participants into clusters based on similar responses. Participants with the same colour belong to the same cluster and share similar characteristics.

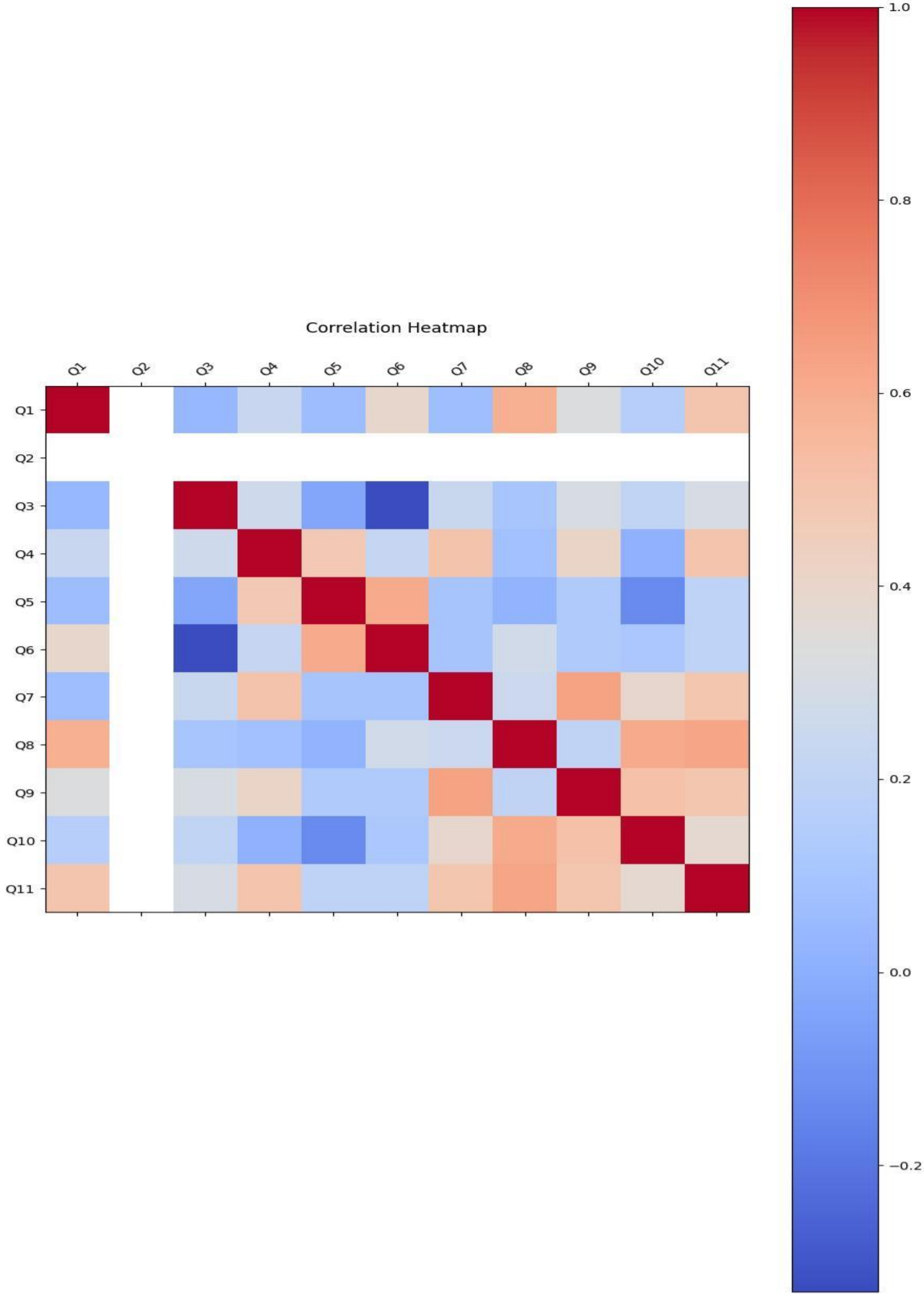
1. **Euclidean Distance bar chart**



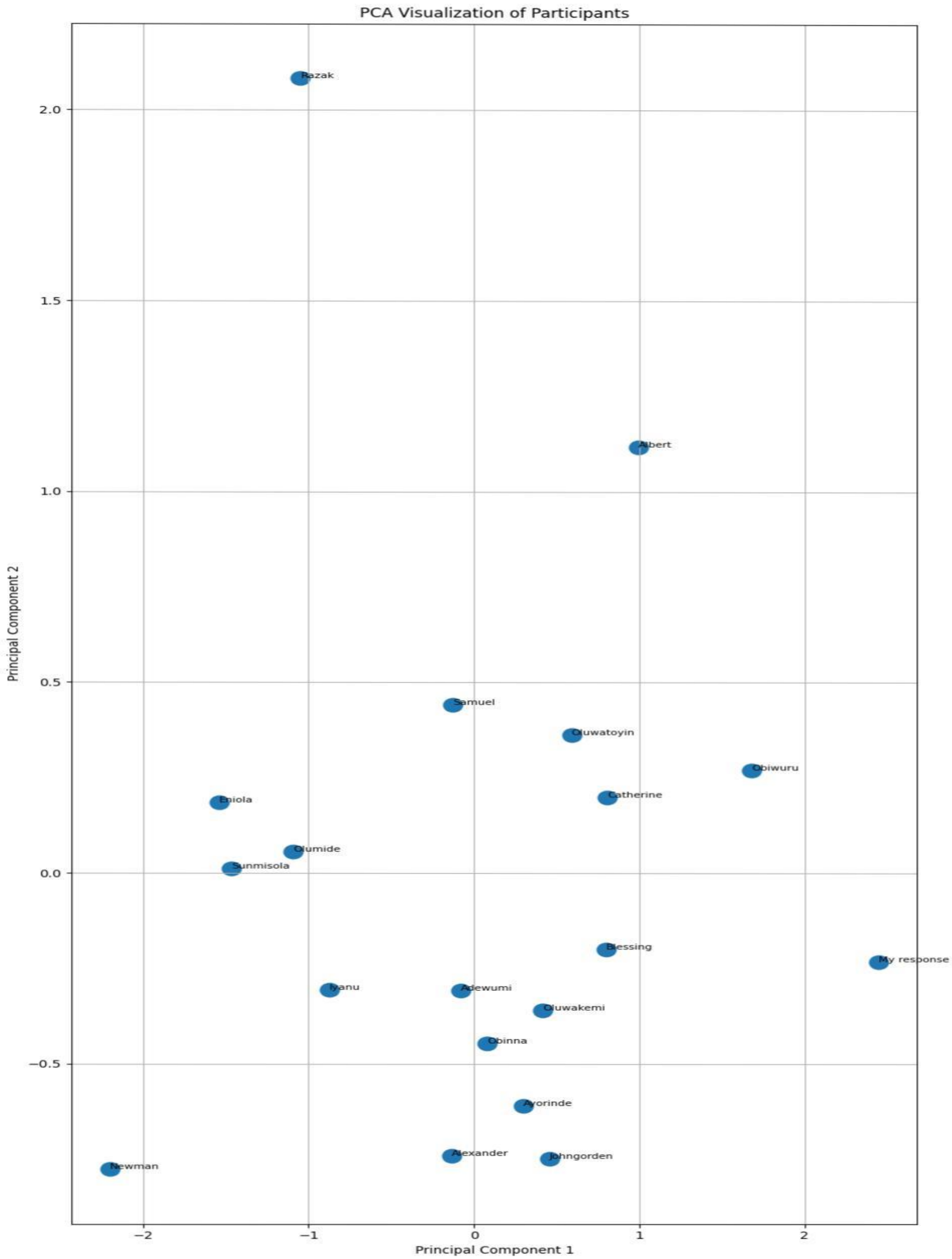
2. Friendship Network Graph



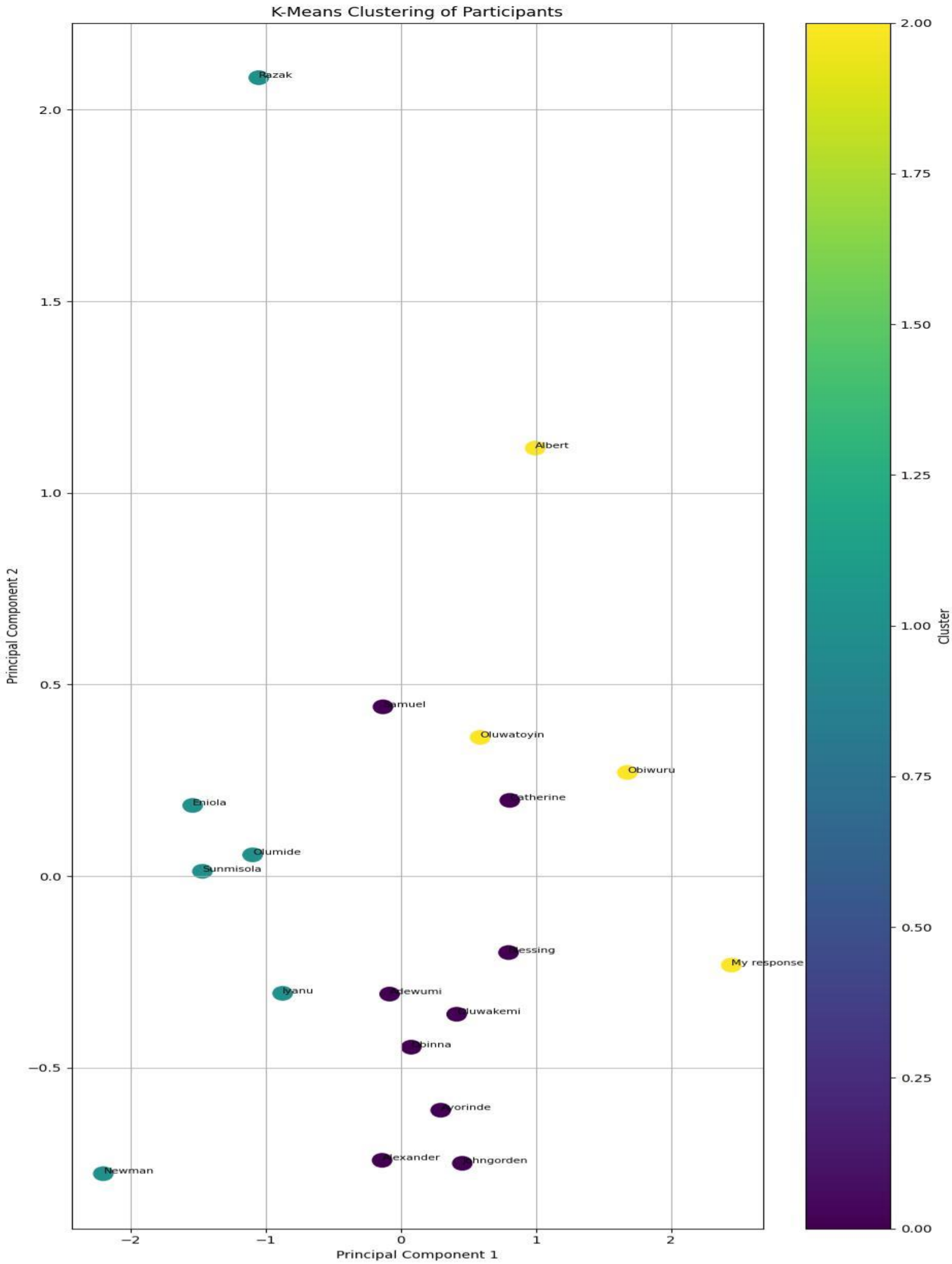
3. Correlation heatmap



4. PCA Visualization



5. K Means Clustering



Conclusion

The project successfully demonstrated how KNN and Euclidean distance can be used to analyse questionnaire responses and identify similar participants. Supporting visualizations provided additional insight into relationships and clusters within the dataset.